RCF Section 1 — Open Expansion, Fracture, and the Reconciliation Propagator
Rewritten Canonical Form — v2.0 (Draft)

Preamble — What This Section Contains and Why

Section 0 established the algebraic foundation (A, admissibility, GNS), the algebraic definitions of $\hat{M}$ and $\hat{F}$, and named the two structural co-primitives — causality (grounded in the Lie algebra A_L) and locality (grounded in the Jordan algebra A_J). These primitives are operationally inert until activated by the Open Expansion Principle, which is the subject of this section.

The v1.0 merge placed the Reconciliation Propagator R_t (§0.4), the Convergence Theorem (§0.5), the thin/full split (§0.6–0.7), the reduced algebra and Dirac bracket (§0.7b), and the Reconciliation Principle (§0.8) all within Section 0 — before causality and locality were named, and before the fracture machinery existed. This created structural problems: $\hat{M}_{red}$ was referenced but never defined (it requires the fracture and the Tier 1/Tier 2 split), Π_{net} forward-referenced K_ω from §2, and the Reconciliation Principle was stated before burden's physical interpretation was available.

This section assembles all of these objects in their correct emergence order. The key sequence is: Open Expansion activates the two primitives → finite rate creates a Lie-Jordan mismatch → the mismatch fractures H_{kin} into sub-Hilbert spaces → $\hat{F}$ gets its physical interpretation (fracture measure) → burden becomes the cost of the fracture → the Tier 1/Tier 2 split becomes visible → A_{red} and $\hat{M}_{red}$ are defined → R_t is constructed → the Convergence Theorem makes Master-Zero a derived result. Every object is defined before use; the chain is acyclic.

Terminology note: The v1.0 terms "SOE" (Single Open Extension) and "MOE" (Multiple Open Extension) are replaced. "SOE" is renamed **Open Expansion** — the local, isometric expansion of locality-built sectors. "MOE" is renamed **Open Extension** — the global, contractive extension of reconciliation across the accumulated expansions. The rename reflects the corrected picture: sub-sectors do not have separate causal layers; they all share the single causal layer and expand to reconcile with it. "Single Open Extension" implied discrete, isolated extension events; "Open Expansion" captures the continuous, driven expansion of sectors toward reconciliation.

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§1.0 Purpose: Activating the Two Primitives

Section 0 named causality and locality as structural co-primitives, grounded in the Lie and Jordan algebras respectively. But these primitives are merely structural possibilities until a dynamic principle activates them. That principle is the Open Expansion Principle: local sectors — built from the Jordan structure (locality) — must expand to reconcile with the single causal layer — built from the Lie structure (causality) — to preserve zero-constraint compatibility.

This activation produces the entire dynamical architecture of the framework:
- The finite rate of reconciliation → the speed limit $c = \gamma \cdot \ell_0$
- The Lie-Jordan mismatch → the fracture of H_{kin} into sub-Hilbert spaces
- The fracture → $\hat{F}$'s physical interpretation (the measure of the mismatch)
- The fracture cost → burden $B_\Delta = \mathrm{Tr}(\rho \hat{F})$
- The fracture structure → the Tier 1/Tier 2 split → A_{red}, $\hat{M}_{red}$, the Dirac bracket
- The dynamics on the fractured structure → R_t = Open Expansion ∘ Open Extension ∘ dephasing
- The asymptotic behavior of R_t → the Convergence Theorem (Master-Zero as derived)
- The physical sector → A_{phy}^{thin}, A_{phy}^{full}, zero-preserving events, operational $\prec$

Every object that was prematurely placed in v1.0 Section 0 is defined here, in the order made possible by the fracture machinery.

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§1.1 Open Expansion and the Finite Rate

The Open Expansion Principle

The two structural primitives — causality (Lie, the "when"-capacity) and locality (Jordan, the "where"-capacity) — are algebraically independent (Theorem 0.4.4). But zero-constraint compatibility (Definition 0.2.2) requires them to be *jointly* consistent: the full product $AB = \frac{1}{2}[A,B] + A \circ B$ requires both the Lie and Jordan parts to be compatible. This joint consistency is not automatic; it must be actively maintained.

The Open Expansion Principle states that local sectors — structures built from the Jordan algebra (locality) — continuously expand their algebraic scope to incorporate new causal data from the single global causal layer (Lie structure, causality), in order to preserve zero-constraint compatibility. This expansion is not an external event imposed on the sectors; it is the internal dynamic consequence of having two independent primitives that must be jointly consistent.

> **POSTULATE 1.1.1 — Open Expansion Principle** [Structural]
>
> Let A be the kinematic algebra with Lie structure A_L (causality) and Jordan structure A_J (locality). The joint Lie-Jordan compatibility required by zero-constraint admissibility is maintained dynamically: local sectors (Jordan-built structures within H_{kin}) continuously expand their algebraic scope to reconcile with the single global causal layer (Lie structure). This expansion is the fundamental dynamic principle of the framework — the mechanism by which the two primitives are jointly activated.

The Finite Propagation Rate

The Open Expansion Principle requires sectors to update their Jordan structure (locality) to remain compatible with the evolving Lie structure (causality). This update cannot occur instantaneously: each incremental expansion requires a finite algebraic check (the zero-constraint compatibility condition must hold at every intermediate step). This imposes a universal maximum rate.

> **THEOREM 1.1.2 — Finite Propagation Rate** [Conditional]
>
> The Open Expansion of any local sector is bounded above by a maximum rate γ, the Open Expansion rate. The corresponding maximum propagation speed is
>
> $c = \gamma \cdot \ell_0$ (1.1.2)
>
> where ℓ_0 is the fundamental length scale of the exact emergent metric (derived in Section 3 from the spectral discreteness of $\hat{F}$).
>
> *Proof sketch:* Each incremental expansion of a sector requires verifying zero-constraint compatibility (Definition 0.2.2) on the new algebraic scope. This verification is a finite algebraic operation requiring time at least $\varepsilon = 1/\gamma$. The corresponding spatial propagation rate is $\gamma \cdot \ell_0$. □
>
> *Status:* [Conditional]. The derivation of γ from deeper principles (e.g., the spectral gap of $\hat{F}$ on $\ker(\hat{M})$) is Theorem Target T-1. At this stage, γ is a primitive parameter of the propagator.

The Lie-Jordan Mismatch

Because Open Expansion propagates at the finite rate γ, the Jordan structure (locality) of any sector cannot instantaneously match the current Lie structure (causality). At any instant, there is a gap between:
- The current Jordan structure — the sector's current co-occurrence relations
- The target Jordan structure — the co-occurrence relations that would be perfectly compatible with the current Lie structure

This gap is the **Lie-Jordan mismatch**. It is the algebraic signature of the finite-rate reconciliation process.

> **DEFINITION 1.1.3 — Lie-Jordan Mismatch** [Structural]
>
> For any local sector with current Jordan structure J_current and target Jordan structure J_target (the Jordan structure compatible with the current Lie structure L), the Lie-Jordan mismatch is the discrepancy
>
> $\delta J := J_{target} - J_{current}$ (1.1.3)
>
> The mismatch is nonzero whenever the sector's Open Expansion has not yet caught up with the causal layer's current state. It vanishes only when the sector has fully reconciled — achieved local Lie-Jordan compatibility.

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§1.2 The Fracture

Fracture of the Hilbert Space

The Lie-Jordan mismatch is not uniformly distributed across H_kin. Different regions of the algebraic structure reconcile at different rates, depending on their local constraint structure and their distance from the causal frontier. As the mismatch accumulates, H_kin fractures into sub-Hilbert spaces — sectors where the Jordan structure has locally converged toward compatibility, even though globally the convergence is incomplete.

> **DEFINITION 1.2.1 — Fracture** [Structural; formal construction conditional on T-2]
>
> The Lie-Jordan mismatch fractures the kinematic Hilbert space H_kin into a collection of sub-Hilbert spaces {H_k}:
>
> H_kin → ⊕_k H_k   (1.2.1)
>
> Each H_k is a sector where the Jordan structure has locally converged toward Lie-Jordan compatibility. The fracture is not an external imposition; it is the algebraic consequence of finite-rate reconciliation producing locally-converged but globally-desynchronized regions.
>
> Correspondingly, the algebra A fractures into sub-algebras {A_k}, each governing the corresponding sub-Hilbert space H_k. Each A_k inherits both Lie and Jordan structure from A, but the Lie-Jordan compatibility is achieved locally within each A_k, not globally across the full algebra.

Remark 1.2.2 — Local Convergence, Global Desynchronization. Each sub-sector H_k converges locally toward Lie-Jordan compatibility — its internal dynamics drives the sector's Jordan structure toward compatibility with its local projection of the causal layer. However, because reconciliation propagates at the finite rate γ, the local compatibilities of distinct sub-sectors are **not synchronized globally**. What one sector "sees" of another sector's state is a lagged snapshot — the other sector's locally-achieved compatibility at the time of the last reconciliation signal to arrive, not its current state.

This local-convergence / global-desynchronization structure is the mechanism that produces the burden (§1.3) and, downstream, the dark energy signature (Section 6). The framework does not require global synchronization; it requires only local convergence within each sector, with the global state being a desynchronized composite.

Remark 1.2.3 — Relation to the "Now". The local Lie-Jordan compatibility achieved within each sector is the precursor to what the framework will call the "now" — the joint definition of "where" (relational/correlational links, constructed in Section 3) and "when" (duration, constructed in Section 4). At this stage, the "now" cannot be defined because neither space nor time exists operationally. What exists is the local Lie-Jordan compatibility — the algebraic precondition for the now. The now itself, and its global desynchronization, are formally constructed in Section 4, after both constituents are available.

F̂'s Physical Interpretation

With the fracture defined, the fracture operator $\hat{F}$ (algebraically defined at §0.3.6) receives its physical interpretation.

> **THEOREM 1.2.4 — $\hat{F}$ as Fracture Measure** [Established (interpretation)]
>
> The fracture operator $\hat{F} = \Sigma_{\{\alpha,\beta\}} \pi_{kin}(\Delta_{\{\alpha\beta\}})^\dagger \pi_{kin}(\Delta_{\{\alpha\beta\}})$ measures the Lie-Jordan mismatch — the algebraic signature of the fracture. Specifically:
>
> (i) $\hat{F}$ is built from the compatibility brackets $\Delta_{\{\alpha\beta\}} = [\hat{C}_\alpha, \hat{C}_\beta]$ — the Lie obstructions of the constraint algebra. These are nonzero precisely where the constraint algebra's Lie and Jordan structures fail to be jointly compatible.
>
> (ii) The expectation value $B_\Delta[\rho] = \mathrm{Tr}(\rho\ \hat{F})$ measures the total Lie-Jordan mismatch in the state ρ — the aggregate fracture across all sectors.
>
> (iii) $\ker(\hat{M}) \subseteq \ker(\hat{F})$ (algebraically provable at §0): if all constraints vanish, all commutators vanish, so $\hat{F}$ vanishes. The reverse inclusion $\ker(\hat{F}) \subseteq \ker(\hat{M})$ is established by the Convergence Theorem (§1.6).
>
> $\hat{F}$ is now physically interpreted as the fracture measure. The quantity $B_\Delta[\rho] = \mathrm{Tr}(\rho\ \hat{F})$, previously an algebraic non-negative number, is now named **burden** — the cost of the fracture, the measure of reconciliation resistance.

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§1.3 Burden

Definition and Physical Meaning

> **DEFINITION 1.3.1 — Burden** [Established (physical interpretation)]
>
> For any density matrix ρ on H_{kin}, the obstruction burden is
>
> $B_\Delta[\rho] = \mathrm{Tr}(\rho \cdot \hat{F}) \in \mathbb{R}_{\{\geq 0\}}$ (1.3.1)
>
> Burden is the cost of the Lie-Jordan mismatch — the measure of reconciliation resistance across all sectors. It is:
> - **Non-negative:** $B_\Delta[\rho] \geq 0$ for all ρ ($\hat{F}$ is positive).
> - **Zero on the physical sector:** $B_\Delta[\rho] = 0$ when ρ is supported on $\ker(\hat{F}) = \ker(\hat{M})$ (the fully reconciled state).
> - **Extensive:** the total burden of a collection of sectors is the sum of individual burdens plus the cross-sector burden (the cost of maintaining coherence across sector boundaries).

Burden Linearity

> **PROPERTY 1.3.2 — Burden Linearity** [Established (proven identity)]
>

> B_Δ is LINEAR in ρ:
>
> $B_\Delta[\Sigma_k p_k \rho_k] = \Sigma_k p_k \cdot B_\Delta[\rho_k]$ (1.3.2)
>
> This is a proven algebraic identity (Tr is linear, $\hat{F}$ is fixed), not a probabilistic average. It is the load-bearing property that licenses the FIREWALL guardrail (Section 2): burden is an algebraic functional evaluated on the full state, not an averaging over outcomes.

Remark 1.3.3 — Burden ≠ Probability. Burden linearity (Property 1.3.2) is structurally distinct from probabilistic averaging. The identity $\mathrm{Tr}(\rho_{kin} \hat{F}) = \Sigma_k p_k \mathrm{Tr}(\rho_k \hat{F})$ holds because Tr is linear and $\hat{F}$ is fixed — it is an algebraic identity, not a measurement postulate. This distinction is what prevents the framework from smuggling probability into gravitational sourcing (the FIREWALL, formally stated in Section 2).

Burden and the Fracture

Burden is the measurable quantity associated with the fracture. It is:
- **Locally small** within each sector that has converged toward Lie-Jordan compatibility
- **Globally nonzero** because the sectors are desynchronized — the cross-sector burden (the cost of maintaining coherence across sector boundaries with mismatched Lie-Jordan states) is nonzero
- **Driven toward zero** by the Reconciliation Propagator R_t (§1.5), which drives each sector toward local Master-Zero compatibility

The burden is the quantity that the Reconciliation Principle (Section 2) will minimize. It is also the quantity from which mass ($m \equiv B_0$, Section 2), gravity ($\Theta^{(B)}_{\mu\nu}$, Section 4), and dark energy ($\rho_{DE} \sim (H/\Gamma)^2$, Section 6) are derived.

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§1.4 The Tier 1 / Tier 2 Separation and the Reduced Algebra

The Compatibility Matrix Reveals the Tiers

The compatibility matrix $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$ (Definition 0.2.4) was introduced at §0 as an algebraic object. With the fracture machinery now available, Δ receives its physical interpretation: it reveals which constraints are **second-class** (Tier 1 — their commutators do not close modulo the constraints; they must be quotiented) and which are **first-class** (Tier 2 — their commutators close; they survive on the reduced algebra).

> **DEFINITION 1.4.1 — Tier 1 / Tier 2 Separation** [Established]
>
> The primitive constraint family $\{\hat{C}_\alpha\}$ separates into:
>
> - **Tier 1 (second-class):** $\{\hat{K}_a\}_{a \in I_1}$ — constraints whose compatibility matrix $\Delta^{(1)}_{ab} = [\hat{K}_a, \hat{K}_b]$ is invertible (non-degenerate). These are the constraints that generate the fracture's Lie obstructions and must be quotiented.
>

> - **Tier 2 (first-class):** $\{\hat{D}_i\}_{i \in I_2}$ — constraints whose commutators close modulo the constraints ($[\hat{D}_i, \hat{D}_j] = f_{ij}{}^k \hat{D}_k$ + Tier 1 terms). These survive on the reduced algebra.
>
> The separation is determined by the rank structure of the full compatibility matrix Δ.

The Reduced Algebra

> **DEFINITION 1.4.2 — Reduced Algebra** [Established]
>
> The Tier 1 ideal $I_K = \langle \hat{K}_a : a \in I_1 \rangle$ is the two-sided (*)-ideal generated by the Tier 1 constraints. The reduced algebra is the quotient
>
> $A_{red} = A / I_K$ (1.4.2)
>
> A_{red} carries the Tier 2 constraints $\{\hat{D}_i\}$ as its surviving constraint structure. Both Lie and Jordan structure descend to A_{red} (the Tier 1 obstructions are quotiented out; the Tier 2 Lie structure closes; the Jordan structure is inherited).

$\hat{M}_{red}$: Separation, Not Reduction

The master constraint $\hat{M}$ (Definition 0.1.6) was characterized as a *collective* — a weighted sum over all constraint ideals (Remark 0.1.7). The Tier 1 / Tier 2 separation splits this collective into two sub-collectives. The Tier 1 sub-collective is quotiented away with I_K. The Tier 2 sub-collective survives on A_{red} as $\hat{M}_{red}$.

> **DEFINITION 1.4.3 — $\hat{M}_{red}$ (Tier 2 Sub-Collective)** [Established]
>
> The reduced master constraint $\hat{M}_{red}$ is the Tier 2 sub-collective of $\hat{M}$:
>
> $\hat{M}_{red} = \Sigma_{\{i \in I_2\}} w_i \cdot \hat{D}_i^\dagger \hat{D}_i \in A_{red}$ (1.4.3)
>
> $\hat{M}_{red}$ is NOT a "reduction" of $\hat{M}$; it is the SEPARATION of $\hat{M}$ into its Tier 1 and Tier 2 sub-collectives, with the Tier 1 sub-collective quotiented and the Tier 2 sub-collective retained on A_{red}. The full $\hat{M} = \hat{M}_{Tier1} + \hat{M}_{red}$; on A_{red}, only $\hat{M}_{red}$ survives.

Remark 1.4.4 — $\hat{M}$ vs. $\hat{M}_{red}$. The relationship between $\hat{M}$ (on A) and $\hat{M}_{red}$ (on A_{red}) is:
- $\ker(\hat{M}) \subseteq A$ is the physical sector of the full algebra — the target of the Convergence Theorem on H_{kin}.
- $\ker(\hat{M}_{red}) \subseteq A_{red}$ is the physical sector of the reduced algebra — the admissibility condition for fields (Section 2).
- The relationship $\ker(\hat{M}_{red}) = \ker(\hat{M}) \cap A_{red}$ holds when the Tier 1 quotient is clean (the Tier 1 constraints are genuinely second-class, not partially first-class). This is verified by the Dirac bracket construction below.

The Dirac Bracket

> **DEFINITION 1.4.5 — Algebraic Dirac Bracket** [Established]
>
> Let $\{\hat{K}_a\}_{a \in I_1}$ be the Tier 1 constraints, with invertible compatibility matrix $\Delta^{(1)}_{ab} = [\hat{K}_a, \hat{K}_b]$. The algebraic Dirac bracket of $A, B \in A$ is
>
> $[A, B]_D = [A, B] - \Sigma_{a,b} [A, \hat{K}_a] \cdot (\Delta^{(1)})^{ab} \cdot [\hat{K}_b, B]$ (1.4.5)
>
> where $(\Delta^{(1)})^{ab}$ is the inverse of $\Delta^{(1)}_{ab}$. The Dirac bracket annihilates Tier 1 constraints: $[A, \hat{K}_c]_D = 0$ for all A and all $c \in I_1$. Equipped with $[\cdot,\cdot]_D$ as its internal commutator, $A_{red} = A / I_K$ is the reduced algebra.

> **THEOREM 1.4.6 — Emergent Dirac Bracket** [Theorem Target T-2]
>
> The Dirac bracket $[\cdot,\cdot]_D$ is the fixed-point effective commutator of the Open Extension component of R_t. Operators that survive global burden minimization satisfy $[A, B]_{eff} = [A, B]_D$.
>
> *Mechanism:* Within-sector Open Expansion generates dressed constraint representatives $\hat{K}_a$; cross-sector Open Extension ensures they are globally valid; the fixed-point commutator coincides with the algebraic Dirac bracket.
>
> *Status:* [Theorem Target T-2]. The full proof requires the stable-mode assumption (spectral gap of R_t on $\ker(\hat{M})$).

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§1.5 The Reconciliation Propagator R_t

The Three-Mechanism Decomposition

The Reconciliation Propagator R_t is the dynamical engine of the framework. It is not a single flow but the ordered composition of three scale-distinct mechanisms: Open Expansion (local, isometric), Open Extension (global, contractive), and dephasing (residual).

> **DEFINITION 1.5.1 — Reconciliation Propagator** [Conditional + T-2]
>
> The Reconciliation Propagator is
>
> $R_t = \lim_{N\to\infty} (R_{T/N}^{OExt} \circ (R_\varepsilon^{OExp})^N)$, $T = N\cdot\varepsilon$ (1.5.1)
>
> where:
> - **R_ε^{OExp}** (Open Expansion, was SOE): the local, isometric expansion of sectors. Propagates ρ under a single algebraic perturbation $\delta\hat{C}$ at rate γ. Isometric / volume-preserving; does NOT minimize total burden. Preserves all quantum phase and spectral structure. Driven by the Lie structure's evolution.

> - **R_{T/N}^{OExt}** (Open Extension, was MOE): the global, contractive extension of reconciliation across accumulated expansions. Bures-metric gradient descent on B_Δ[ρ]. Contractive, irreversible, burden-minimizing. Drives the Jordan structure toward compatibility with the Lie structure. The Master-Zero condition is enforced as a Lagrange multiplier, not postulated.
> - **Dephasing**: the residual suppression of cross-eigenspace coherences, the leading-order signature of Open Extension on the F̂ spectral decomposition.

The Open Expansion Flow (Local, Isometric)

> **DEFINITION 1.5.2 — Open Expansion Flow** [Conditional]
>
> The Open Expansion flow propagates ρ under a single algebraic perturbation δĈ to the constraint set. The perturbation is defined entirely within Layer L/Q (no reference to events or causal order — breaking the v1.0 dependency loop):
>
> (i) Flux-Gradient Flow (local constraint alignment):
> dρ/dt|_{OExp}^{flux} = ∇ · J_Δ(ρ) = Σ_{αβ} [Ĉ_α, [Δ_{αβ}, ρ Ĉ_β†]] + h.c.   (1.5.2a)
>
> (ii) Spectral-Gradient Flow (eigenbasis alignment):
> dρ/dt|_{OExp}^{spec} = i [√(F̂ + δF̂(δĈ)), ρ]   (1.5.2b)
> where δF̂(δĈ) = [δĈ, [δĈ, F̂]] + O(δĈ^3)
>
> Properties: Isometric / volume-preserving. Preserves Tr(ρ M̂) and Tr(ρ F̂) locally — burden is REDISTRIBUTED, not erased. Preserves all quantum phase and spectral structure. δĈ is purely algebraic — no Section 1+ vocabulary required.

The Open Extension Flow (Global, Contractive)

> **DEFINITION 1.5.3 — Open Extension Flow** [Conditional + T-2]
>
> Over a chain of N ≥ 2 Open Expansions (duration T = N·ε), the Open Extension flow is the Bures-metric gradient descent:
>
> dρ/dt|_{OExt} = − η · ∇_{Bures} B_Δ[ρ] − λ_t · ∇_{Bures} C[ρ]   (1.5.3a)
>
> where:
> - ∇_{Bures} = functional gradient w.r.t. the Bures metric
> - B_Δ[ρ] = Tr(ρ F̂) (obstruction burden)
> - C[ρ] = Tr(ρ M̂) (Master-Zero target, enforced as Lagrange multiplier; NOT postulated)
>
> Derived Lindblad form (Theorem Target T-2): at leading order for large N, the effect on cross-sub-sector coherences is
>
> dρ/dt|_{OExt}^{lead} = − Γ [F̂, [F̂, ρ]]   (1.5.3b)
>

> with relaxation rate $\Gamma \propto N\cdot\eta$ and spectral-gap dependence $(f_i - f_j)^2$.

The Dephasing Residual

> **DEFINITION 1.5.4 — Dephasing** [Conditional]
>
> Between $\hat{F}$-eigenspaces with distinct eigenvalues:
>
> $d\rho/dt|_{deph} = -\Gamma [\hat{F}, [\hat{F}, \rho]]$ (1.5.4)
>
> Suppresses residual cross-eigenspace coherences while preserving within-eigenspace structure. Note: equations (1.5.3b) and (1.5.4) coincide at leading order — dephasing is the leading-order signature of Open Extension on the $\hat{F}$ spectral decomposition.

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§1.6 The Convergence Theorem — Master-Zero as Derived Result

The Central Theorem

The Convergence Theorem is the central result of Section 1. It states that the Reconciliation Propagator drives any unconstrained kinematic state asymptotically toward the physical sector — the kernel of $\hat{M}$. This makes Master-Zero a *derived* result, not a postulate.

> **THEOREM 1.6.1 — Convergence to the Physical Sector** [Conditional; T-2 for ∞-dim]
>
> Let ω_{kin} be ANY positive state on A (not pre-constrained). Under the composed Open Expansion–Open Extension flow R_t (Definition 1.5.1),
>
> $\lim_{t\to\infty} \rho_t = \hat{P}_0 \rho_0 \hat{P}_0 / \mathrm{Tr}(\hat{P}_0 \rho_0)$ (1.6.1)
>
> where $\hat{P}_0$ projects onto $\ker(\hat{M}) = \ker(\hat{F})$.
>
> **Key consequences:**
> 1. $\mathrm{Tr}(\rho_\infty \hat{M}) = 0$ — Master-Zero is asymptotically DERIVED, not assumed.
> 2. $\mathrm{Tr}(\rho_\infty \hat{F}) = 0$ — the obstruction burden is fully resolved (locally, within each sector).
> 3. Cross-sub-sector coherences decay exponentially with rate $\Gamma \cdot (\Delta f_{ij})^2$.
>
> *Mechanism assignment:*
> - Open Expansion incorporates new perturbations without increasing global burden (isometric).
> - Open Extension drives global convergence to $\ker(\hat{M})$ (contractive).
> - Dephasing suppresses residual cross-eigenspace coherence.
>
> *Proof sketch.* Open Expansion is isometric (preserves spectral decomposition of $\hat{F}$). Open Extension is contractive on the Bures metric and drives $B_\Delta[\rho]$ monotonically downward. By the LaSalle invariance principle applied to the Bures-gradient flow, the ω-limit

set is contained in the critical set of B_Δ, which (by the reverse kernel inclusion proved below) coincides with $\ker(\hat{F}) = \ker(\hat{M})$. □
>
> *Status:* [Conditional Theorem]. The finite-dimensional toy model ($M_6(\mathbb{C})$, non-commuting constraints, $\mathrm{Tr}(\rho_0 \hat{M}) = 0.96 \to 2.14\times10^{-4}$ in finite t) verifies convergence numerically. The infinite-dimensional proof requires spectral gap conditions on $\hat{F}$ (Theorem Target T-2).

Master-Zero Equivalence

> **COROLLARY 1.6.2 — Master-Zero Equivalence** [Established on the physical sector]
>
> On the asymptotic physical sector, the following are equivalent:
>
> (i) $\omega_\infty(\hat{M}) = 0$
> (ii) $\omega_\infty(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$ for all α
> (iii) $\omega_\infty(\hat{F}) = 0$
> (iv) $\omega_\infty(\Delta_{\alpha\beta}^\dagger \Delta_{\alpha\beta}) = 0$ for all α, β
>
> In particular, $\ker(\hat{M}) = \ker(\hat{F})$ on the physical sector, completing the reverse inclusion of the algebraic kernel containment (§0).

Local Convergence, Global Desynchronization

> **THEOREM 1.6.3 — Local Convergence, Global Desynchronization** [Structural; formal proof conditional on T-2]
>
> The Convergence Theorem holds LOCALLY within each sub-sector H_k: each sector's internal dynamics drives its state toward local $\ker(\hat{M})$ compatibility. However, because Open Expansion propagates at the finite rate γ, the local convergences of distinct sectors are NOT synchronized globally.
>
> The global state ρ_∞ is a desynchronized composite of locally-converged sectors:
>
> $\rho_\infty = \oplus_k\, p_k \cdot \rho_\infty^{(k)} + \rho_{cross}$ (1.6.3)
>
> where $\rho_\infty^{(k)}$ is the locally-converged state of sector k, p_k are the sector weights, and ρ_{cross} is the cross-sector coherence (suppressed by dephasing but not identically zero due to ongoing Open Expansion).
>
> The residual cross-sector burden — the cost of maintaining this desynchronized composite — is the quantity that, at cosmological scale (Section 6), appears as dark energy.

$\Lambda_B = 0$ (Cosmological Constant Vanishes)

> **COROLLARY 1.6.4 — $\Lambda_B = 0$** [Conditional on T-2]
>

> The burden cosmological term Λ_B vanishes identically as a consequence of Master-Zero being an exact asymptotic attractor:
>
> $\Lambda_B = 0$ (1.6.4)
>
> This is not a fine-tuning; it is a structural consequence of the asymptotic attractor. The formal action S_{eff} (Section 2) carries a Λ_B term that vanishes by Master-Zero, eliminating the cosmological constant problem at the algebraic level (subject to closure of T-2).

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§1.7 The Thin/Full Split of A_{phy}

The Thin Candidate (Available Early)

The thin physical sub-algebra is defined using only $\ker(\hat{M})$-compatibility — available immediately after the Convergence Theorem.

> **DEFINITION 1.7.1 — Thin Physical Sub-Algebra** [Established]
>
> The thin physical sub-algebra is the set of algebra elements whose kinematic representation preserves $\ker(\hat{M})$:
>
> $A_{phy}^{thin} = \{ A \in A : \pi_{kin}(A) \ker(\hat{M}) \subseteq \ker(\hat{M}) \}$ (1.7.1)
>
> Equivalently: $A \in A_{phy}^{thin}$ iff $[A, \hat{P}_0] = 0$ on $\ker(\hat{M})$, where $\hat{P}_0$ is the asymptotic projector onto $\ker(\hat{M})$.
>
> Properties: Defined using only Layer Q objects (A, π_{kin}, $\ker(\hat{M})$). Does NOT require fixed-point closure under R_∞. Does NOT require the Reconciliation Principle. Sufficient for defining zero-preserving events (§1.8) without forward reference.

> **LEMMA 1.7.2 — Thin Candidate is a Sub-Algebra** [Established]
>
> A_{phy}^{thin} is a unital (*)-sub-algebra of A.
>
> *Proof.* Standard closure verification (addition, multiplication, involution, unit). □

The Full Algebra (Available Late)

> **DEFINITION 1.7.3 — Full Physical Sub-Algebra** [Established]
>
> The full physical sub-algebra is the fixed-point algebra of the asymptotic Reconciliation Propagator $R_\infty = \lim_{t\to\infty} R_t$:
>
> $A_{phy}^{full} = \{ A \in A : [A, R_\infty] = 0 \}$ (1.7.3)
>

> Equivalently: A ∈ A_phy^full iff A commutes with P̂_0 AND preserves the full F̂-spectral decomposition of ker(M̂) (not just the kernel itself).

Theorem 1.7.4 — Fixed-Point Characterization (Thin = Full)

> **THEOREM 1.7.4 — Fixed-Point Characterization** [Theorem Target T-2]
>
> Under the stable-mode assumption on the reconciliation flow:
>
> A_phy^thin = A_phy^full   (1.7.4)
>
> *Proof sketch.*
> (⊆) If [A, R_∞] = 0 then A commutes with P̂_0, hence A ∈ A_phy^thin.
> (⊇) If A ∈ A_phy^thin, the stable-mode assumption implies A's action on the ker(M̂) decomposition is invariant under R_t's flow. By Master-Zero equality ker(M̂) = ker(F̂), this extends to R_∞, giving [A, R_∞] = 0. □
>
> *Status:* [Theorem Target T-2]. The stable-mode assumption is the same one used in Theorem 1.4.6 (Dirac bracket) and is the framework's central technical conjecture.

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§1.8 Zero-Preserving Events and the Operational Causal Order

With A_phy^thin available, causality (named as a primitive at §0.4.1, grounded in the Lie algebra) receives its operational definition.

Zero-Preserving Events

> **DEFINITION 1.8.1 — Zero-Preserving Event** [Established]
>
> A zero-preserving event is an element E ∈ A_phy^thin that preserves the physical sector:
>
> E ∈ A_phy^thin such that [E, P̂_0] = 0 as an operator on H_kin   (1.8.1)
>
> (equivalently, E preserves the full P̂_0-eigenspace decomposition, not just ker(M̂)).
>
> The set of all zero-preserving events is denoted E_phy.

Remark 1.8.2 — Events and the Lie Structure. Events are elements of A_phy^thin, which inherits both Lie and Jordan structure from A. The Lie structure of events — [E_1, E_2] — is what the operational causal order (below) is built from. The Jordan structure — E_1 ∘ E_2 — is what the correlational structure (Section 3) is built from. The two are independent, as established at §0.

The Operational Causal Order

> **DEFINITION 1.8.3 — Primitive Causal Relation** [Established (definition); Theorem 1.8.4 for partial-order properties]
>
> For zero-preserving events E_1, E_2 ∈ E_phy, the primitive causal relation is:
>
> E_1 ≺ E_2 iff (i) d(E_1) < d(E_2) (depth inequality)
>
                   (ii) π_kin(E_2) π_kin(E_1) |ψ⟩ ≠ 0 for some |ψ⟩ ∈ ker(M̂) (relational connectivity)   (1.8.3)
>
> where d(E) is the reconciliation depth (the number of Open Expansion steps required to incorporate E into the reconciled sector).

> **THEOREM 1.8.4 — Causal Partial Order** [Conditional (requires Assumption 1.8.5)]
>
> The relation ≺ is a strict partial order on E_phy: irreflexive, asymmetric, transitive.
>
> *Proof sketch.* Irreflexivity and asymmetry follow from the depth inequality (i). Transitivity requires the relational connectivity (ii) to compose — this is Assumption 1.8.5. □

> **ASSUMPTION 1.8.5 — Necessity Composition** [Structural assumption]
>
> If E_1 ≺ E_2 (E_1 is necessary for E_2) and E_2 ≺ E_3 (E_2 is necessary for E_3), then E_1 is necessary for E_3 — the relational connectivity composes across the causal chain.
>
> This is the structural analogue of transitivity of logical entailment, applied to the reconciliation structure. It is plausible but not derived from the algebra; it is an explicit assumption.

The Operational "When"-Capacity

With the causal order ≺ operationally defined, causality (named as a primitive at §0.4.1) now has its operational form. The Lie structure — the algebraic substrate of causality — is realized as the strict partial order ≺ on zero-preserving events. This is the operational "when"-capacity: the dependency ordering of events.

Remark 1.8.6 — Duration Is Still Deferred. The causal order ≺ is the operational form of causality, but it is NOT yet duration (the actual "when"). Duration — burden-weighted causal depth — requires the burden (available, §1.3) and the causal depth (available, §1.8.3), but its construction as a physical quantity (proper time) is deferred to Section 4. At this stage, ≺ is the ordering; duration is the measure on that ordering.

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§1.9 Architectural Summary

What Section 1 Contains

Object	Definition	Status
Open Expansion Principle	Post 1.1.1	Structural
Finite propagation rate $c = \gamma \cdot \ell_0$	Thm 1.1.2	Conditional (T-1 for γ derivation)
Lie-Jordan mismatch	Def 1.1.3	Structural
Fracture of H_{kin} into $\{H_k\}$	Def 1.2.1	Structural (T-2 for formal construction)
$\hat{F}$ physical interpretation (fracture measure)	Thm 1.2.4	Established (interpretation)
Burden $B_\Delta = \mathrm{Tr}(\rho \hat{F})$	Def 1.3.1	Established
Burden linearity	Prop 1.3.2	Established (proven identity)
Tier 1 / Tier 2 separation	Def 1.4.1	Established
Reduced algebra $A_{red} = A / I_K$	Def 1.4.2	Established
$\hat{M}_{red}$ (Tier 2 sub-collective)	Def 1.4.3	Established
Dirac bracket	Def 1.4.5	Established (algebraic); Thm 1.4.6 (dynamical, T-2)
Reconciliation Propagator R_t	Def 1.5.1	Conditional + T-2
Open Expansion flow (was SOE)	Def 1.5.2	Conditional
Open Extension flow (was MOE)	Def 1.5.3	Conditional + T-2
Dephasing	Def 1.5.4	Conditional
Convergence Theorem (Master-Zero derived)	Thm 1.6.1	Conditional (T-2 for ∞-dim)
Master-Zero equivalence	Cor 1.6.2	Established (on physical sector)
Local convergence, global desynchronization	Thm 1.6.3	Structural (T-2 for formal proof)
$\Lambda_B = 0$	Cor 1.6.4	Conditional on T-2
Thin physical sub-algebra A_{phy}^{thin}	Def 1.7.1	Established
Full physical sub-algebra A_{phy}^{full}	Def 1.7.3	Established
Thin = Full (Thm 1.7.4)	Thm 1.7.4	Theorem Target T-2
Zero-preserving events E_{phy}	Def 1.8.1	Established
Operational causal order $\prec$	Def 1.8.3 + Thm 1.8.4	Conditional (Assumption 1.8.5)

What Section 1 Does NOT Contain (Deferred)

Object	Deferred to	Reason
Reconciliation Principle (variational)	§2	Needs burden (available) + correlational structure (§3) for full target
Dark energy mechanism ($\rho_{DE} \sim (H/\Gamma)^2$)	§2 (introduction) + §6 (application)	Needs Γ evolution argument
Fields, particles, mass-burden identity	§2	Needs A_{red}, $\hat{M}_{red}$, events (all available)
Record sectors, Born rule, FIREWALL	§2	Needs fracture + events
Correlation kernel K_ω, relational/correlational links	§3	Needs GNS state + sectoral structure
Duration (burden-weighted causal depth)	§4	Needs $\prec$ + burden
The "now" (joint where+when)	§4	Needs space (§3) + time (§4)
Gravity, burden tensor	§4	Needs burden + coarse-grained metric
Three-channel burden decomposition	§2 or §4	Needs stable modes (§2)

The Emergence Ladder Through Section 1

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From §0:
  A (algebra) → Lie-Jordan decomposition → M̂ (collective) → admissibility
  → GNS → H_kin → F̂ (algebraic)
  + Causality (Lie primitive) + Locality (Jordan primitive)

§1:
  Open Expansion Principle → finite rate c = γ·ℓ₀
  → Lie-Jordan mismatch → fracture (H_kin → {H_k})
  → F̂ physical interpretation (fracture measure)
  → Burden B_Δ = Tr(ρ F̂) (cost of the fracture)
  → Tier 1/Tier 2 split → A_red → M̂_red (separation, not reduction)
  → Dirac bracket
  → R_t = Open Expansion ◦ Open Extension ◦ dephasing
  → Convergence Theorem (Master-Zero DERIVED, not postulated)
  → Local convergence, global desynchronization
  → Λ_B = 0
  → A_phy^thin / A_phy^full (thin = full under T-2)
  → Zero-preserving events E_phy
  → Operational causal order ⋖ (causality activated)

→ Section 2: RP, dark energy mechanism, fields, particles, mass, Born rule
→ Section 3: K_ω, correlational links, space (operational "where")
→ Section 4: duration (operational "when"), the "now", gravity
```

### What This Section Unlocks for Section 2

With Section 1 closed, Section 2 can be merged against a stable dynamical foundation. Section 2 will introduce:

1. **The Reconciliation Principle** (variational) — the system evolves under $R_t$ to minimize total relational inconsistency $I(S)$, where $S$ is a set of mutually correlated events. The variational target uses events (§1.8), $\lessdot$ (§1.8), and correlation strength $s(a,b)$ (introduced algebraically, with full geometric interpretation in §3).
2. **Dark energy mechanism** — $\rho_{DE} \sim (H/\Gamma)^2$ with $\Gamma \sim \eta \cdot B_\Delta$. The restorative drive of Open Extension, with the $\Gamma$ evolution argument (larger in the past when burden was higher) resolving the early dark energy tension.
3. **Fields** — effective reconciliation dynamics on $A_{red}$, preserving $\omega(\hat{M}_{red}) = 0$.
4. **Field modes and particles** — stable modes with bounded burden; particles as low-burden stable modes.
5. **Mass-burden identity** — $m \equiv B_0$, resolving the §3.2.5 forward reference.
6. **Record sectors and decoherence** — stable record separation from $B_{cross} \gg B_{intra}$.
7. **Born rule** — $p_i = |c_i|^2$ from Z-envariance as Open Extension fixed-point symmetry.
8. **FIREWALL** — the formal guardrail separating probabilistic branch weights (Layer A) from algebraic burden (Layers A/B/C), grounded in burden linearity (Property 1.3.2).

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Acyclicity Test

Question: Does Section 1 define every object before using it? Are there forward references to §2+ objects?

Answer: The chain is acyclic within Section 1.

- Open Expansion (§1.1) uses only the two named primitives from §0 (causality/Lie, locality/Jordan) and the zero-constraint compatibility (§0.2.2).
- The fracture (§1.2) uses the Lie-Jordan mismatch (§1.1) and F̂ (algebraic, §0.3.6).
- F̂'s physical interpretation (§1.2.4) uses the fracture (§1.2) and F̂ (§0.3.6).
- Burden (§1.3) uses F̂ (§0.3.6) and the fracture interpretation (§1.2.4).
- The Tier 1/Tier 2 split (§1.4) uses the compatibility matrix Δ (§0.2.4) and the fracture structure.
- A_red (§1.4.2) uses the Tier 1 ideal (§1.4.1).
- M̂_red (§1.4.3) uses M̂ (§0.1.6) and the Tier 1/Tier 2 split (§1.4.1). It is a SEPARATION of the collective, not a new object — no circularity.
- The Dirac bracket (§1.4.5) uses the Tier 1 compatibility matrix (§0.2.4, §1.4.1).
- R_t (§1.5) uses F̂ (§0.3.6), M̂ (§0.1.6), and the fracture structure (§1.2). δĈ is algebraic (§0).
- The Convergence Theorem (§1.6) uses R_t (§1.5), F̂ (§0.3.6), M̂ (§0.1.6).
- The thin/full split (§1.7) uses ker(M̂) (§1.6) and R_∞ (§1.5).
- Events (§1.8) use A_phy^thin (§1.7).
- The causal order < (§1.8) uses events (§1.8.1) and ker(M̂) (§1.6).

No forward references to §2+ objects. The Reconciliation Principle, K_ω, duration, the "now," gravity, and the three-channel decomposition are all deferred. The correlation strength s(a,b) needed for the RP's variational target is introduced in §2 (as an algebraic precursor to K_ω), not here.

One residual concern: The cross-extension network operator Π_net (v1.0 Def 0.3.8) is still not defined. It requires K_ω (§3) and the sectoral decomposition (§1.2). It is deferred to §3 or introduced in §2 as needed. The relational burden channel T^(rel) = [Ĉ_α, Π_net] (the dark matter mechanism, §5) depends on this operator. This is flagged as a forward reference but not a circularity — Π_net will be defined when its ingredients are available.

Verdict: Section 1 is acyclic. The architecture holds through the dynamical foundation.

End of Section 1 — Rewritten Canonical Form v2.0 (Draft).

This draft assembles the fracture machinery in emergence order. The key things to evaluate:

Does M̂_red work? Definition 1.4.3 defines M̂_red as the Tier 2 sub-collective of M̂, not a "reduction." This is the resolution to the v1.0 gap where M̂_red was referenced but never defined. The construction works because the Tier 1/Tier 2 split (made visible by the fracture and the compatibility matrix) separates M̂ into two sub-collectives, with the Tier 1 sub-collective quotiented and the Tier 2 sub-collective surviving on A_red. No circularity.

Does the desynchronization mechanism hold? Theorem 1.6.3 states the local-convergence / global-desynchronization structure. Each sector converges locally to ker(M̂) compatibility; finite propagation prevents global synchronization; the global state is a desynchronized composite. This is the mechanism that produces dark energy (§6) — the residual cross-sector burden is the cost of desynchronization. The "now" interpretation is properly deferred to §4.

Does the SOE→Open Expansion / MOE→Open Extension rename work? The rename reflects the corrected picture: sectors expand (Open Expansion, local, isometric) and reconciliation extends across the accumulated expansions (Open Extension, global, contractive). The mathematics is unchanged; the physical interpretation is sharpened.

The Π_net residual. The cross-extension network operator is still deferred — it needs K_ω (§3). This is flagged honestly. If the relational burden channel (dark matter mechanism) is needed before §3, Π_net will need an algebraic precursor introduced earlier, similar to how s(a,b) is introduced as a precursor to K_ω in §2.